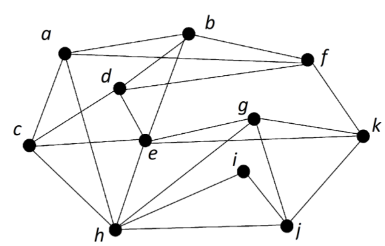
MATH120-24S2 Discrete Mathematics

Assignment 10

Due: 11:59pm Friday 11 October
Submit a single PDF document (1 file only) to the submission link in Learn. It is your responsibility to ensure that your submission is clear and easily readable, and that you upload the correct file by the due date.

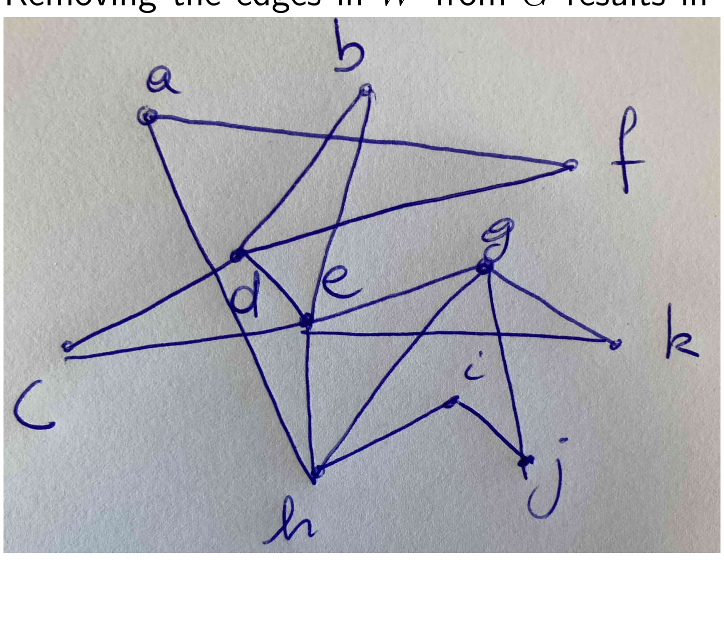
1. Run EULERCIRCUIT algorithm (from Tutorial 10) to find an Eulerian circuit of the following graph G. Include your intermediate steps.



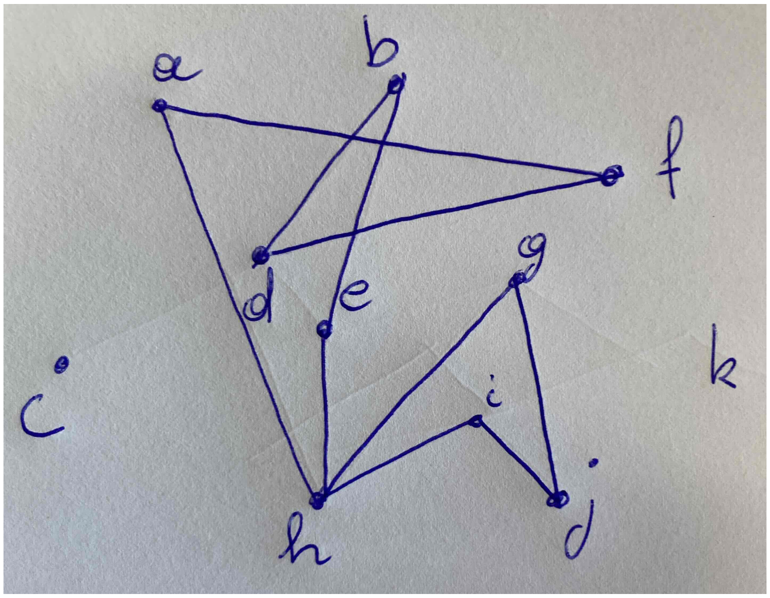
Solution:

(a) We start with the closed walk $W = abfkjhca$

(b) Removing the edges in W from G results in the following graph:



(c) Let $W_1 = cdegkec$, then the resulting graph is:



(d) Let $W_2 = bdfahgjiheb$.

(e) Finally, we take W and replace c with W_1 and b with W_2 which results in the Eulerian circuit: $abdfahgjihebfkjhcdegkeca$.

2. (a) From the information given below about a graph, determine whether it is Hamiltonian, not Hamiltonian, or if you don't have enough information.

   i. A simple graph G_a has 8 vertices of degrees 4, 4, 4, 4, 5, 5, 5 and 6.

   ii. G_c is a connected graph on 8 vertices where every vertex has degree 2

   (b) We asked ChatGPT to produce a graph that contained a bridge and is Hamiltonian. Consider its answer below and explain why it is incorrect.

   An example of a graph that contains a bridge and is Hamiltonian can be constructed by considering a scenario where the graph is connected by a single edge between two Hamiltonian subgraphs. Despite the presence of the bridge, a Hamiltonian cycle can still exist.

   Example: *Consider a graph $\mathcal{G}$ made of two complete graphs K_3 and K_4, connected by a single edge e between one vertex in K_3 and one vertex in K_4.*

   The graph $\mathcal{G}$:

- *Vertices:* $\{A, B, C\}$ *for* K_3 *and* $\{D, E, F, G\}$ *for* K_4.
- *Edges in* K_3*:* $\{(A, B), (A, C), (B, C)\}$
- *Edges in* K_4*:* $\{(D, E), (D, F), (D, G), (E, F), (E, G), (F, G)\}$
- *Bridge: the edge* $e = (C, D)$

A Hamiltonian cycle could be: $(A, B, C, D, E, F, G, D, C, A)$*, which visits each vertex exactly once and returns to the starting vertex.*

Solution:

(a) i. G_a is simple and has $n = 8$ vertices where each vertex has degree at least $4 = n/2$. G_a satisfies Dirac's Theorem and is therefore Hamiltonian. (Note however, that the graph cannot exist as the sum of its degrees is odd, this was not intentional)

ii. G_c must be a cycle on 8 vertices and is therefore Hamiltonian.

(b) The supposed "Hamilton cycle" proposed by ChatGPT repeats the bridge (C, D) so it is not a cycle and therefore not a Hamilton cycle.

3. *You do not need to simplify exponents but you must show your working.*

(a) How many (unordered) pairs of two distinct integers chosen from the set $\{1, 2, 3, \ldots, 101\}$ have a sum that is even.

(b) Poker: In a poker hand (randomly chosen set of 5 cards from standard 52-card deck):

i. There are $\binom{52}{5}$ poker hands; how many of those consist of two pairs and a different card?

ii. How many ways are there to get neither a repeated denomination, nor five adjacent denominations.

Notes: cards are *adjacent* if they are consecutive in the ordering A, 2, 3,..., 10, J, Q, K, A, 2, ...
A *denomination* is a number (e.g. 5) or letter (J, Q, K, A).
A *pair* is two cards of the same denomination.

Solution:

(a) Two integers have a sum that is even if they are either both even or both odd. There are 50 even integers in the set and 51 odd integers. Therefore there are $\binom{51}{2}$ ways to select two odd integers and $\binom{50}{2}$ ways to select two even integers. Therefore there are $\binom{50}{2} + \binom{51}{2} = \frac{50*49}{2} + \frac{51*50}{2} = 25(49+51) = 2500$ such pairs.

(b) i. There are $\binom{13}{2}$ ways to obtain the two denominations for the pairs, then 11 choices for the denomination of the single card. For the suits of the cards in the pair, we have $\binom{4}{2}$ possibilities, while for the single card, we have 4 possibilities. Hence there are

$$\binom{13}{2}\binom{4}{2} \cdot \binom{4}{2} \cdot 11 \cdot 4 = 13 \cdot 12 \cdot \frac{1}{2} \cdot 6^2 \cdot 11 \cdot 4 = 123,552$$

poker hands with a this combination (a "two pair").

ii. $(\binom{13}{5} - 13) \times 4^5$.

There are $(\binom{13}{5} - 13)$ ways to choose the denominations. This is because there are $\binom{13}{5}$ ways to choose 5 distinct denominations, then we remove the 13 ways that these could be 5 consecutive denominations.

Then there are 4^5 ways to choose the suits for these cards.